Pranav M

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Education

2023 - 2027 Electronics and Communication Engineering at **PES University**
2022 Class 12th at **Vydehi School of Excellence**

Projects

Aletheia GitHub
Built a processing-near-memory (PNM) framework on a Raspberry Pi 5, dispatching commands over TCP to the Pi as a memory node instead of moving data; ran PMU-backed characterization (working-set sweeps, stride scans, dataset scaling) of memory access patterns.

Aetheron GitHub
Implemented a RISC-V SoC at simulation level with TileLink-lite interconnect and peripherals including UART and GPIO timers using Bluespec.

Herald GitHub
Taped out a DSP coprocessor via Tiny Tapeout featuring MAC and CORDIC engines (4 operations each) for accelerating math on FPU-less microcontrollers.

Chorion GitHub
Implemented ChaCha20 stream cipher accelerator on PYNQ-Z2 FPGA (Xilinx Zynq-7000 SoC) for high-speed cryptographic operations.

Experience

HSP, PES University — Mentor Jul 2025 – Present
Mentored students on digital hardware design projects across two consecutive terms, guiding architecture design, reviewing HDL, and helping debug design modules.

Skills

Languages	SystemVerilog, Verilog, Bluespec, C, Python, Assembly
Tools & Platforms	Vivado, GTKWave, Vitis IDE, Verilator, ModelSim, Git, VS Code, Neovim, Linux
Domains of Interest	Memory Systems, DDR/Memory Interfaces, SoC Design, CPU Design, Performance Analysis